

Shaibal Muhtadee

647-526-1088 | shaibalmuhtadee@gmail.com | linkedin.com/in/shaibalmuhtadee | shaibalmuhtadee.com

EDUCATION

University of Toronto

Bachelor of Applied Science in Computer Engineering

Toronto, ON

Sept 2020 – Apr 2025

EXPERIENCE

Software Engineering Intern

Zebra Technologies

May 2023 – Sept 2024

Toronto, ON

- Built authentication and authorization services in Rust to secure API access for over 200 autonomous robots
- Developed TypeScript APIs for securely storing and retrieving API keys, JWTs, and robot configuration data
- Implemented attribute-based access control with a custom made policy parser, allowing teams to update authorization rules without modifying application code
- Built internal interfaces using React and Flutter for managing secrets, access policies, and robot configurations
- Created a Rust integration test suite with 99% code coverage, validating authentication flows, API behaviour, and service communication
- Developed a CLI that retrieved and connected secrets from GitHub, GCP, and AWS for development workflows
- Containerized services with Docker and supported CI/CD deployments to Kubernetes environments

PROJECTS

Inokta | *Independent Startup Project, Next.js, FastAPI, TypeScript, Python*

June 2025 – June 2026

- Co-developed an AI workspace that transforms uploaded documents into an editable world model of entities, attributes, and relationships
- Built Alice, a LangGraph-based AI harness that extracts entities from documents and generates persistent text, table, and graph modules
- Implemented document ingestion and retrieval pipelines supporting PDF extraction, table detection, text chunking, vector embeddings, and semantic search with ChromaDB
- Developed an interactive interface for creating, editing, and connecting entities, allowing users to correct and extend relationships inferred from source documents
- Implemented streamed AI responses and tool activity over WebSockets, with session-scoped conversations and Model Context Protocol tool integration

GanttWise | *Python, React, Node.js, MongoDB*

Sept 2024 – Apr 2025

- Built dynamic web app using React and TypeScript to automate task planning with Deep Reinforcement Learning
- Used Python scripts to preprocess datasets and debug learning agents, optimizing scheduling under constraints
- Integrated a full-stack pipeline with Node.js backend and MongoDB, enabling real-time project updates

Searchington | *Python, Flask, Redis, AWS*

Sept 2024 – Dec 2024

- Built a web search engine with a crawler that recursively visits URLs, extracts HTML, creates word lists, and stores data in SQLite, using an inverted index and various metrics to deliver relevant search results
- Hosted on AWS and implemented caching with Redis to optimize query performance, improving response times by over 70%

ChromaMap | *C++, OpenStreetMap API, GTK Library, Git*

Sept 2022 – Apr 2023

- Created a Geographic Information System mapping software using C++ and the OpenStreetMap API
- Implemented Dijkstra algorithm and A* pathfinding heuristics to solve the travelling salesman problem
- Improved browse and load times by over 300% by developing a Level of Detail tiling scheme for the map
- Optimized application speed by multi-threading and code visualization, stack trace and flame graphs

TECHNICAL SKILLS

Languages: TypeScript, Python, Rust, C++, SQL, HTML/CSS, Bash

Frameworks: React, Next.js, FastAPI, Node.js, Flask, Flutter, LangGraph

Developer Tools: Docker, Kubernetes, AWS, GCP, Linux, Git, CI/CD

Libraries: MongoDB, Redis, ChromaDB, SQLite, REST APIs, WebSockets